NorthWind Traders Analysis

1. Project Overview and Objective

This project analyses Northwind Traders' order and customer information to understand sales activity, customer distribution and product performance. Excel is used for data preparation and Power BI is used for data modelling, DAX calculations and interactive reporting.

The main objective is to convert raw business data into a clean, structured and interactive report that helps identify sales patterns, compare performance across categories and locations, and support business decisions through data-driven insights.

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2. Data Sources

- **Source Description and Timeline:** Maven Analytics **and** 2013-2015.
- **Domain**: Sales Analytics

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3. Problem Statement

- To analyse order and sales performance across products, categories and customer locations to identify major contributors and trends.
- To examine customer distribution by country and city and understand the geographical spread of the customer base.
- To analyse shipping-related information to understand the cost of delivering customer orders.
- To identify changes in order activity over time and provide practical recommendations for improving sales and customer service.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
CustomerID	Text	Unique code identifying each customer.
CompanyName	Text	Name of the customer company.
ContactName	Text	Name of the primary contact person.
ContactTitle	Text / Categorical	Job title of the customer contact.
City	Text / Categorical	City where the customer is located.
Country	Text / Categorical	Country where the customer is located.
CatergoryID	Numerical	Catergory id

CategoryName	Text / Categorical	Catergory
Description	Text	Description about category
EmployeeId	Numerical	Employee id
EmployeeName	Text	Name of the employee's
Title	Text/categorical	Role/ position of Employee
City	Text/categorical	Location of Employee
Country	Text	Location of Employee
Reports To	Text	To Whom they Reports
ProductId	Numerical	ProductID
ProductName	Text	Name of the product
QuantityPer unit	Text	How many quantities
UnitPrice	Numerical	Price of each quantity
Discontinued	Numerical	Discontinuation of product
CategoryId	Numerical	Identifier linking a Categories to a Product.
ShipperId	Numerical	Shipper Id
Company name	Text/Categorical	Company name of shipper
OrderID	Text / Whole Number	Unique identifier for an order in the Orders table.
OrderDate	Date	Date on which the customer order was placed.
RequiredDate	Date	Date requested by the customer for delivery.
ShippedDate	Date	Date on which the order was shipped.
Freight	Decimal / Currency	Shipping and handling cost recorded for the order.
ProductID	Text / Whole Number	Identifier linking an order detail to a product.
Quantity	Whole Number	Number of units ordered.
Unit Price	Decimal / Currency	Price per unit of the product.
Discount	Decimal Number	Discount applied to the order detail, if present.
OrderId	Numerical	Identifier linking an order detail to Northwind trader

ProductId	Numerical	Identifier linking an Product detail to Northwind trader
Unti price	Numerical	Price of each quantity
Quantity	Numerical	Number of quantity
Discount	Numerical	Discount value
Total amount	Numerical	Calculated total amount

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query) (Sample)

Tasks Performed:

Data Cleaning & Transformation: Inspected the customer data, identified five duplicate records in the uncleaned practice sheet, checked blank fields, and standardised inconsistent text such as company names, contact names, contact titles, city and country values.

Filtering & Sorting: Used filters and sorting to review customer records and organise data for analysis.

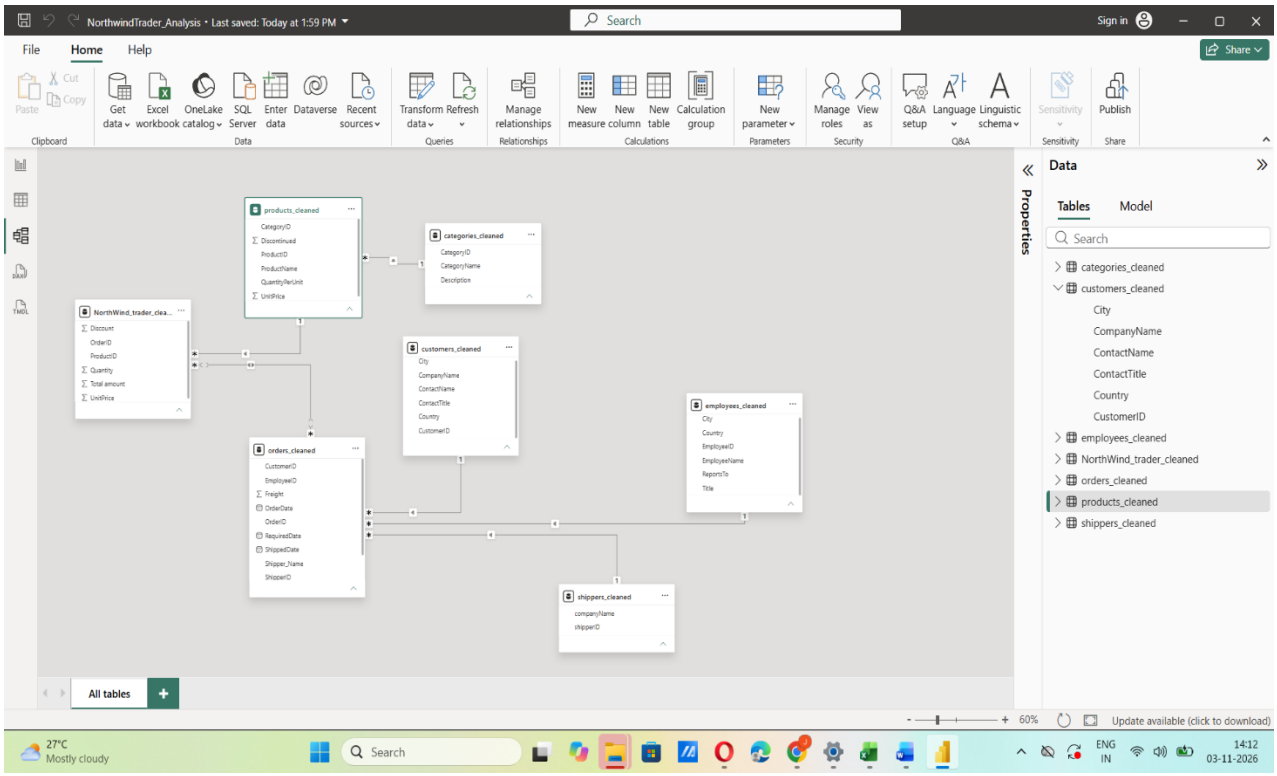
Data Validation: Removed Duplicate data**,** filled null values with meaningful answer

Data Structure: Organised order transactions as the central fact data and customer, product, category and employee details as descriptive lookup/dimension data where these tables are available in the Power BI model.

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- **Calculated Columns & DAX Measures:**
  - Create Measure to calculate Gross Sale.
  - Created Measure to calculate Discount amount
  - Created Measure to calculate Net Sales
  - Created Measure to calculate Distinct Total order
  - Created Measure to calculate Total Quantity
  - Created Measure to calculate Average Discount value
  - Created Measure to calculate Average Order value
  - Created Measure to calculate Not Shipped orders
  - Created Measure to calculate Delayed order
  - Created Measure to calculate on time order
  - Created Measure to calculate for Sales YTD
  - Created column to visualize only year from order date for slicer and charts.

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8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:**
  - Created Text box for heading as North Wind trader analysis.
  - Added 8 Cards that visualize **Net Sales, Total order, Total Customer, Average order value, Sales YTD, on time order, delayed order, Not shipped order**.

- Created a cluster Bar Chart to represent sales by Product name.
- Created donut chart to visualize Net sales by Category
- Created line chart to Visualize Net Sales by Year
- Created Stacked column Chart to visualize Net sales by country
- Created table chart for Drill through to visualize sales overview by category with net sales, total quantity, discount amount, gross sales in a table format.
- Created a tree map chart to visualize net sales by category
- Created scatter chart to visualize average discount and net sales by product
- Created stacked bar chart to visualize delayed orders by shipper.
- Created Map chart to visualize net sales by city and country.
- Created Funnel chart to visualize net sales by Employee name.
- Created a stacked bar chart to visualize total orders by title.
- Created slicer to visual Year wise sales trend.
- Added Bookmark and modified into button to shift from 1 dashboard to another
- Created Slicers to visualize Category wise.

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9. Insights & Conclusions

Key Findings:

- Analysed overall business performance through sales, orders, customers, and shipping KPIs.
- Identified sales contributions across products, categories, countries, and employees.
- Observed yearly sales trends and differences in shipping performance.
- Examined discount patterns across products.

1. Descriptive Analysis – What happened?

- KPI Cards: Summarize Net Sales, Total Orders, Total Customers, Average Order Value, and shipping status.
- Donut Chart: Shows Net Sales by Category.
- Clustered Bar Chart: Displays Net Sales by Product Name.
- Stacked Column Chart: Represents Net Sales by Country.
- Map Chart: Shows geographical sales distribution.
- Funnel Chart: Displays Net Sales by Employee.
- Line Chart: Presents yearly Net Sales trends.
- Stacked Bar Chart: Shows Total Orders by Employee Title.

2. Diagnostic Analysis – Why did it happen?

- Scatter Chart: Examines the relationship between Average Discount and Net Sales by Product

- Delayed Orders by Shipper: Helps compare shipping delays across different shippers.
- Drillthrough Table: Provides detailed category-wise sales, quantity, and discount information.

3. Predictive Analysis – What might happen next?

- Line Chart – Historical yearly sales trends to identify possible future growth or decline.

4. Prescriptive Analysis – What should be done?

- Delayed Orders by Shipper – Review shipping performance and reduce delays.
- Sales by Product and Category – Prioritize products and categories with stronger sales contributions.
- Sales by Country and City – Identify locations requiring business attention.

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10. Conclusions

The Northwind Traders Analysis project provided practical experience in data cleaning, transformation, data modelling, and visualization using Excel, Power Query, and Power BI. I identified and handled missing values, duplicates, inconsistent formats, and data quality issues. I created relationships between tables and developed DAX measures to analyse key business metrics.

An interactive dashboard was designed using various charts, KPI cards, slicers, drill through, and bookmarks. The dashboard helped analyse sales trends, product performance, customer activity, employee contributions, and shipping efficiency.

Overall, this project enhanced my data analytics skills and helped me understand how to convert raw data into meaningful business insights.